What a Public Log Reveals: Observing Play Style in Literature (Canadian Fish)

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Abstract

In the `us54` dialect of Literature (Canadian Fish), every action is broadcast: each ask and its result, each declaration with the true holders of all six cards, and the public hand counts. This paper studies the *observation* problem in isolation from the adaptation problem treated by its companion: given only that public channel, how much of a seat's play style can be recovered from a single game, and what fundamentally limits the recovery? We specify the channel exactly — including what it certifies (a hit locates a card; an ask certifies a licence; a declaration reveals its true holders, making foreign and own-hand-only declares *exact* public observations, not inferences) and what it structurally omits (declining to declare emits no event, so declare-window patience is invisible to any log-based observer). From the channel we build a 14-dimensional feature vector of rates and shares per seat, calibrate a deliberately simple diagonal-Gaussian classifier on 1,350 mirror games, and report two small empirical methods findings: the Gaussian normaliser term must be dropped (it handed one style 40 of 60 control-game reads through a calibration-tightness bonus), and per-style variances must be shrunk halfway to pooled. In the calibrated observable space the roster's geometry is stark: four of the nine styles form a quadrangle nearly at a single point (pairwise pooled-z distance 0.20–0.25 — the companion papers' inert-axis finding arriving in observable space), while the Turtle sits 3.2–3.7 from that cluster. Measured on 1,800 held-out cross-play games, single-game top-1 accuracy over nine styles rises from 16.4% at 40 events to 22.4% at the full log against an 11.1% chance rate — twice chance, and no better than that — with per-style accuracy from 50.4% (Ghost) down to 5.6% (Balanced). The ceiling is not the classifier's: styles that diverge from the control on 0.39–2.89% of decisions — roughly one to three policy-revealing decisions per seat per game, some of them invisible declines — upper-bound what *any* observer of this channel can recover in one game. We close by locating this instrument inside the opponent-detection program of the project's research synthesis: which panel statistics this implementation ships, and which remain future work.

1 Introduction

Literature (Canadian Fish) is a six-player, two-team card game in which all hidden information originates at the deal and every subsequent action is public. That makes it an unusually clean domain for a question that is usually confounded: *how much of an opponent's temperament is visible in what they publicly do?* In most imperfect-information games the observer's problem is entangled with hidden state — one cannot tell a bold player from a lucky one without knowing their cards. In Literature the log eventually tells you the cards too: every successful ask moves a named card face up, and every declaration reveals, by rule, the true holder of all six cards it names.

The FishAI project's v0.5 paper [1] measured *which* of nine parameterised play styles wins, holding skill fixed by giving every style one identical full-strength inference engine. Its forthcoming v1.0 companion [2] asks whether an agent that identifies its opponents' styles online and adapts can beat the fixed champion. Between those two questions sits a third, which this paper isolates: the observation problem itself. Before asking whether a read is *useful*, one should measure whether a read is *possible* — and the answer turns out to have enough structure, and enough honest negative content, to warrant its own report.

Concretely, this paper documents the observation layer shipped in the repository as `observe.ts` and `classify.ts` [10]: a single pass over the public log that produces a fixed feature vector per seat, and a diagonal-Gaussian classifier over those vectors calibrated offline from mirror games. Its contributions are:

1. **An exact specification of the observation channel** (Section 2): what the `us54` public log certifies — including the result that foreign declares and own-hand-only declares are *exact* public observations because declarations reveal true holders — and what it structurally cannot show, chiefly that declining to declare emits no event.
2. **A 14-feature behavioural fingerprint** (Section 3), every entry a rate or share, with the measured game-length confound that forced that normalisation.
3. **Two empirical methods findings from calibration** (Section 4): dropping the Gaussian normaliser, and shrinking per-style variances halfway to pooled. Both were made by measurement, not taste, and both are the kind of small correction that vanishes from most write-ups.
4. **The geometry of the roster in observable space** (Section 5): a pooled-z distance matrix computed from the committed fingerprints, in which the inert-axis finding of the companion papers [3, 1] reappears as four styles collapsed onto nearly a single point.
5. **Measured single-game accuracy and its confusion structure** (Section 6): 22.4% top-1 over nine styles against 11.1% chance on held-out cross-play games, with the Ghost and Scout acting as attractors that receive 47.4% of all reads.
6. **The fundamental limit and the program** (Section 7): decision divergence rates of 0.39–2.89% [7, 1] upper-bound any log-based observer, and the instrument is located inside the opponent-detection panel of the project's research synthesis [5], with the shipped and unshipped panel rows stated explicitly.

Throughout, "the classifier" means the committed implementation with the committed calibration [8, 10]; every quantitative claim traces to a committed artifact [9, 8] or to the repository's measurement documents [6, 7].

2 The observation channel

2.1 What the public log contains

The `us54` engine's public log [6, 10] is a sequence of typed events: `game_started`; `ask` with asker, target, the named card, and whether it hit; `claim` (a declaration) with the claimer, the book, the claimer's stated assignments, the *true* holders of all six cards, and the outcome; and the bookkeeping events `pass`, `designate`, `player_out`, `endgame`, `game_over`. Hand counts are public at all times. This is exactly the payload a human at the table would have: bots consume only a `SeatView`, and a test enforces that no hidden hand is reachable [1, 10].

2.2 What the log certifies exactly

The observation layer records only facts the rules force to be true; nothing in the feature extractor is an estimate. Four certifications carry the whole construction [6]:

1. **A hit locates a card.** The named card transfers face up (`RULES_US54.md` row 9), so after a hit the card is publicly at the asker until it moves again or its book resolves. This yields a public card-location map.

2. **An ask certifies a licence.** Holding at least one card of the asked book is the only licence to ask into it (row 6), so every ask proves the asker held such a card at that moment — and row 7 proves the asker *lacked* the named card. The certification is decayed conservatively: it is dropped when the book resolves, when the seat runs out of cards, and when a hit strips a card of that book from the seat, since the stripped card may have been the licence itself.

3. **A declaration reveals its true holders.** The `claim` event carries the actual holder of each of the six cards, whatever the outcome. Consequently a *foreign* declare (the claimer holds no card of the book: the claimer appears nowhere among the true holders) and an *own-hand-only* declare (every true holder is the claimer) are **exact public observations**. These are precisely the public signatures of the `foreignDeclare` and `declareOnlyOwnHand` style mechanisms of the roster [7, 1], and their exactness is what makes the corresponding features noiseless.

4. **Hand counts replay.** Counts are re-derived from the log rather than read from the view: every seat starts at the dealt hand size, a hit moves one card, and a claim removes each of the six cards from whoever the reveal says held it. This is deliberate: the classifier evaluates *truncated* logs (Section 4), and a truncated view's own top-level counts describe the end of the game, not the truncation point. The replayed counts are correct at every prefix, and the test suite pins them against the engine's counts at full length and at mid-game snapshots [10].

2.3 What the log cannot show

Declining to declare emits **no event**: the `us54` declare window advances silently through seats that pass on their option [6, 10]. Declare-window *patience* — how long a seat sat on a declarable book before firing — is therefore invisible to any observer of this channel, not merely to this implementation. No shipped feature depends on it. The feature that comes closest, `declareBackload`, measures where in the *observed event stream* a seat's declares landed (mean declare event index over the final observed index), which is the observable stand-in for the decline-counting "declare move index" a god's-eye harness would use.

This blind spot matters more than it looks, and Section 7 returns to it: some of the very decisions on which styles differ are declines, and those decisions leave no trace at all.

Table 1: The 14 features of the shipped observation vector, in classifier order [10]. All are rates or shares; denominators are floored at 1.

#	Feature	Definition
1	`hitRate`	hits / asks
2	`askDiversity`	distinct books asked into / asks
3	`sameBookRepeatRate`	consecutive same-book asks by the seat / (asks − 1)
4	`certainAskShare`	asks whose named card was publicly located at the target / asks (a guaranteed public hit)
5	`deadAskShare`	asks whose named card was publicly located elsewhere / asks (a guaranteed public miss)
6	`leakyAskShare`	asks into books where the asker's team already publicly accounted for ≥ 4 of six cards (located plus certified, never double-counted) / asks
7	`completionAskShare`	asks into books with five cards publicly located at the asker's team / asks
8	`missFewestShare`	of the seat's misses, the fraction whose target held strictly the fewest cards among live opponents (replayed counts at ask time)
9	`missMostShare`	... strictly the most
10	`askShare`	the seat's asks / all observed events
11	`declareShare`	the seat's declares / all observed events
12	`foreignDeclareShare`	foreign declares / declares (exact; Section 2.2)
13	`ownHandOnlyShare`	own-hand-only declares / declares (exact)
14	`declareBackload`	mean declare event index / final observed event index

3 The feature set

Table 1 lists the vector. Three design decisions deserve their measured justifications.

Shares, never raw counts. Every entry is a rate or a share, length-invariant by construction. This is a correction, not a preference: the first calibration ran over raw counts and read every short cross-play game as "not the style" simply because the mirror games it was calibrated on ran longer — a fact about the *opponents*, not the seat [10]. A count observed at 150 events is incommensurable with a fingerprint calibrated at a different game length; a share projects a 70-event clinch and a 200-event grind into the same space. Checkpoint buckets (Section 4) still matter on top, because shares themselves drift over a game: declares concentrate late, and public certainty accumulates.

Deliberate exclusions. Raw `asks`, `hits` and `declaresCorrect` are absent. The first two are already the denominators inside `hitRate` and `askShare`; correctness is near ceiling for every full-strength style (roster declare precision 0.946–0.990 [1]), so it separates nothing and would only add correlated noise to a diagonal model.

A feature that never fired. In the entire calibration — 8,100 seat-vectors from 1,350 mirror games — `completionAskShare` is identically zero for all nine styles [8]. The interpretation (offered as interpretation) is structural: a full-strength engine whose team publicly holds five cards of a book almost always has the certainty to declare it rather than ask into it. Within this roster the column separates nothing and, under the classifier's variance floor, penalises all styles equally; it is retained because it is aimed at play outside this roster — weaker or human play, where chasing a publicly finished book is a real behaviour. Relatedly, `deadAskShare` is the public signature of the contained-book turn-pass studied in the second companion [4]: that mechanism's deliberate guaranteed misses are exactly what feature 5 counts.

4 Calibration and the classifier

4.1 Fingerprints from mirror games

Fingerprints are generated offline and committed [8]: per style, 150 `us54` mirror games (all six seats the same style), seeds `fingerprint-v1-<style>-<i>`, start seat i mod 6, observed through the *same code path* the classifier uses — the generator and the classifier share the extractor, or the fingerprints would describe a different instrument than the one that reads them [10]. Each game contributes six seat-vectors, so the full-log bucket holds 900 vectors per style.

Because several behaviours drift over a game, each style carries one fingerprint per *checkpoint bucket*: log prefixes of 60, 120, 200 and 300 events, plus `full` (completed games). A bucket only collects games whose log outlived it, and the committed provenance records how quickly the population thins: at 120 events, between 180 (Blitz) and 882 (Turtle) of the 900 seat-vectors remain; only the Turtle's games reach 200 events in any number (246 vectors); *no* game reached 300, so that bucket carries the full-game statistics with a recorded sample count of zero [8]. Bucket selection at read time: a view whose log ends in `game_over` reads `full` — it *is* a completed game, whatever its length — otherwise the nearest checkpoint at or below the observed event count, falling back to 60 below 60 events [10].

4.2 The model, and two corrections made by measurement

The model is deliberately the simplest thing that can be honest: a diagonal Gaussian per style over the 14-vector. For an observed vector x and style s with per-feature calibration mean $\mu_{s,i}$ and standard deviation $\sigma_{s,i}$,

$$\ell_s(x) \;=\; -\tfrac{1}{2}\sum_{i=1}^{14} z_{s,i}^2, \qquad z_{s,i} \;=\; \frac{x_i - \mu_{s,i}}{\tilde{\sigma}_{s,i}},$$

and a softmax over the nine ℓ_s gives the raw posterior. Diagonal on purpose: with 14 features and a few hundred vectors per bucket, a full covariance would be an estimate of noise, and accuracy is *measured* downstream (Section 6), never assumed here. Two modelling choices were forced by calibration failures [10]:

1. **The Gaussian normaliser is dropped.** The textbook log-likelihood carries a $-\ln\sigma_{s,i}$ term per feature. With sample SDs over a few hundred vectors, $\sum_i -\ln\sigma_{s,i}$ is a large *constant* bonus for whichever style happened to calibrate tightest — and in the first calibration it handed the Ghost 40 of 60 mirror-Balanced reads. Without the term the score is a pure z-distance, whose argmax at a style's own calibration mean is that style itself. The cost is that the model is no longer a normalised density; the benefit, measured, is that it stops rewarding calibration tightness as if it were evidence.

Table 2: Pairwise distances between the committed full-log fingerprints [8], in pooled-z units: $d(a,b) = (\sum_i((\mu_{a,i} - \mu_{b,i})/\bar{\sigma}_i)^2)^{1/2}$ with $\bar{\sigma}_i$ the RMS of the nine styles' calibration SDs for feature i (floored at 0.01; the floor binds only on the identically zero completion column). Computed directly from the committed table; reproducible with a few lines of `node`.

	Balanced	Blitz	Punter	Banker	Turtle	Hoarder	Scout	Ghost
Blitz	0.22							
Punter	0.24	0.25						
Banker	0.20	0.23	0.24					
Turtle	3.22	3.35	3.18	3.28				
Hoarder	1.74	1.79	1.74	1.76	2.81			
Scout	1.66	1.79	1.64	1.69	2.12	1.97		
Ghost	1.02	1.04	1.15	0.97	3.70	2.15	2.07	
Archivist	0.95	1.08	0.96	0.99	2.58	1.74	0.75	1.42

2. **Per-style SDs are shrunk halfway to pooled:** $\tilde{\sigma}^2_{s,i} = (\sigma^2_{s,i} + \bar{\sigma}^2_i)/2$, where $\bar{\sigma}_i$ is the root-mean-square of the nine styles' SDs for feature i. Raw per-style SDs make the widest-variance style an outlier magnet — everything unusual "must be the Scout" — while fully pooled SDs discard real information (the Turtle genuinely is more repetitive than its mean alone says). The halfway blend beat both endpoints on cross-play calibration probes and tied them on mirror self-reads.

Both corrections are small, empirical, and the kind of thing a methods section usually omits. They are reported here because each one, left unfixed, produces a confidently wrong instrument, and because neither is specific to this game: any few-sample per-class Gaussian over behavioural rates faces both.

4.3 Honesty mechanisms

Two further mechanisms are structural rather than statistical [10]. *Sample-size damping*: the posterior is blended toward uniform by $w = \min(1, \text{asks}/12)$, so at zero observed asks the answer is exactly "I don't know," and full confidence is reachable only once a seat has asked a dozen times — asks, not events, because asks are the seat's own observed choices, and a seat that has barely moved in a long game is still unread. *A variance floor* of 0.01 prevents the one literally-constant calibration column (Section 3) from turning a single stray count into an infinite log-penalty; it becomes a very strong finite vote instead.

The unit-test suite pins the contract rather than the accuracy: the posterior is a proper distribution; zero asks damp to exactly uniform; the bucket rule is pinned point by point; the classifier is deterministic and never mutates a frozen view; and one deliberately loose real-game smoke test asserts only better-than-chance on the roster's most separable axis (Turtle-family reads at roughly 0.31 against a $2/9 \approx 0.22$ chance rate) [10]. Real accuracy is a measured number (Section 6), not a unit-test assertion — a tight bound there would be a number the suite cannot honestly own.

5 Separability: the roster's geometry in observable space

Table 2 is the paper's central picture: the distance between every pair of committed full-log fingerprints, in pooled-z units. Four structures stand out, and each has a mechanism.

The quadrangle. Balanced, Blitz, Punter and Banker sit pairwise 0.20–0.25 apart — over a fourteen-dimensional space, nearly a single point. This is the inert-axis finding of the companions [3, 1] arriving in observable space. Those four styles are exactly the ones whose defining knob was the declare threshold, an axis measured to select identical declares across the roster's entire range; what behavioural difference survives (their ask-scoring weights) amounts to 0.39–1.19% of decisions against the control [7], and $\leq 1.3\%$ of decisions leaves almost no trace in fourteen public rates. The tournament separates these styles by up to 5.4 points of mean score [1]; the public log, in one game, essentially cannot.

The far pole. The Turtle sits 3.2–3.7 from the quadrangle and the Ghost, and no closer than 2.1 to anything — the roster's one style an observer can genuinely expect to spot. Its per-feature signature is broad, not narrow: relative to Balanced, `missFewestShare` $+1.7\bar{\sigma}$, `leakyAskShare` $+1.6\bar{\sigma}$, `deadAskShare` $+1.4\bar{\sigma}$, `hitRate` $-0.9\bar{\sigma}$, `askDiversity` $-0.8\bar{\sigma}$. The mechanism is its own-hand-only declare gate [1]: refusing to bank split books keeps resolved information on the table, so its asks are measured against a position the whole table has already solved.

The Hoarder, and what the hoard-gate fix bought. The Hoarder sits 1.74 from Balanced — and essentially the whole distance is one axis, `foreignDeclareShare` at $+1.5\bar{\sigma}$ (calibration mean 0.225 against the control's 0.009). This is the audited hoard gate's measured side effect: a style that refuses declares which spend its licences ends up declaring disproportionately the books that cost it nothing, i.e. foreign ones [7, 1]. Two facts qualify the 1.74. First, it exists only because of the audit: the first Hoarder implementation was byte-identical to Balanced — zero divergent decisions in 275,380 — so its observable distance was exactly 0 [1]. A style must first *exist* to be observed. Second, the distance is strongly time-dependent: at the 60-event bucket the Hoarder–Balanced distance is 0.15, because declares — the only place its signature lives — concentrate late. Early-game reads of the Hoarder are close to structurally impossible.

Scout $\approx$ Archivist. The two information styles sit 0.75 apart, with no single feature differing by more than $0.44\bar{\sigma}$. The Archivist is the interesting half of the pair: its name-giving move — declaring books it holds no card of, for its teammates — fires at a shipped threshold of 0.95 and almost never triggers, so its measured foreign-declare rate (0.007) is actually *below* the control's (0.014) [1]. Its public signature is therefore not its thesis but its ask-weight side effects, which resemble the Scout's. A style whose defining mechanism rarely fires is, to this channel, mostly its side effects.

6 Measured accuracy

6.1 Design

Accuracy is measured on games the calibration never saw, in a setting deliberately harder than calibration: *cross-play.* All 36 unordered style pairings play 50 single games each (team 0 one style, team 1 the other; seeds `clsacc-v1-<i>`, disjoint from the calibration prefix; start seat rotating), for 1,800 games. Each game is classified from log prefixes of 40, 80, 150 and 250 events and from the full log, using the committed classifier and fingerprints end to end; a truncation is scored only for games whose log outlived it. Each checkpoint therefore yields up to $1{,}800 \times 6 = 10{,}800$ seat-reads, each scored top-1 against the seat's true style among nine

Table 3: Single-game top-1 accuracy over nine styles, by observed-event checkpoint [9]. Chance is $1/9 \approx 0.111$. A checkpoint scores only games whose log outlived it; the 150-event row's small, style-skewed survivor population (112 of 1,800 games, dominated by the long-game styles) is why its higher figure is not comparable to the others, and no game reached 250 events.

Checkpoint (events)	Seat-reads	Top-1	vs. chance
40	10,800	0.164	1.5×
80	10,464	0.202	1.8×
150 (survivors only)	672	0.375	—
full log	10,800	0.224	2.0×

Table 4: Full-log confusion, row-normalised to percent (1,200 seat-reads per row) [9]. Rows are true styles, columns predictions; the diagonal is bold.

True \ read	Bal	Bli	Pun	Ban	Tur	Hoa	Sco	Gho	Arc
Balanced	**5.6**	8.3	15.2	14.4	2.2	4.4	15.3	27.0	7.8
Blitz	5.3	**9.3**	16.6	14.2	1.3	4.8	14.3	26.1	8.2
Punter	4.8	9.1	**12.9**	12.5	3.2	5.2	16.4	27.5	8.4
Banker	5.5	8.5	14.5	**12.6**	2.8	3.4	16.1	32.3	4.3
Turtle	2.2	2.8	4.4	7.7	**27.2**	7.8	23.1	19.3	5.7
Hoarder	1.2	3.7	6.3	9.0	9.3	**30.9**	17.3	18.3	3.9
Scout	2.5	4.1	4.1	4.9	15.2	1.2	**41.3**	16.9	9.9
Ghost	3.2	5.8	11.9	11.4	0.8	2.2	10.3	**50.4**	3.9
Archivist	2.8	4.8	6.0	5.6	12.3	1.9	33.1	21.8	**11.8**

candidates (chance $1/9 \approx 11.1\%$) [9]. The distribution shift from mirror calibration to cross-play deployment is intentional — it is the shift the adaptive engine [2] actually faces — and its cost is visible in the numbers below.

6.2 The accuracy curve

Table 3: accuracy rises from 16.4% at 40 events through 20.2% at 80 to 22.4% on the full log — twice chance, and no better than that. The curve's shape is the damping and the buckets doing their jobs (early reads are pulled toward uniform; later reads have more declares to work with), and its *level* is the separability geometry of Section 5 made concrete. One row needs its caveat stated plainly: the 37.5% at 150 events is computed over the 672 seat-reads of the 112 games that lasted that long, a population heavily skewed toward the long-game styles (Turtle 186 and Scout 180 of the 672 seats), which are also the separable ones. It is a fact about survivors, not a point on the same curve.

6.3 Per-style accuracy and the confusion structure

The headline 22.4% decomposes into two regimes (Table 4). The styles with a real observable signature are read far above chance: Ghost 50.4%, Scout 41.3%, Hoarder 30.9%, Turtle 27.2%. The quadrangle is read at 5.6–12.9% — *at or below* chance for Balanced and Blitz — exactly as the distance table predicts for four styles 0.2 apart under calibration SDs an order of magnitude larger.

This split was pre-stated. The run's committed predictions include (`P4`) that accuracy "should be good for turtle, ghost and hoarder," and heavily confused inside the Balanced–Blitz–

Punter–Banker quadrangle, on the Stage-1B separability read [9]. The recorded verdict is *mixed*, and honestly so: the quadrangle confusion held (mean 10.1% against a $\leq 35\%$ condition), but the separable trio under-shot its $\geq 50\%$ bar (mean 36.2%; only the Ghost cleared it) — and quadrangle errors did *not* mostly stay inside the quadrangle (30–40% against a $\geq 50\%$ condition). Where they went instead is the table's most interesting structure.

Ghost and Scout as attractors. Summing columns: the Ghost is predicted for 2,875 of 10,800 reads (26.6%) and the Scout for 2,245 (20.8%) — together 47.4% of all reads, for two styles that are 2/9 of the truth. Every quadrangle style's single most common read is Ghost (26–32%); the Archivist is read as Scout nearly three times as often as as itself (33.1% vs. 11.8%).

We offer an interpretation, and flag it as interpretation rather than measurement. The per-feature decompositions of Section 5 show the roster's two strongest observable axes are the ones the Ghost and the Scout respectively own: the ask-repetition/hit-rate direction (Ghost: fewer same-book repeats, more certain asks, higher hit rate — leak avoidance's public shadow) and the dead-ask/targeting direction (Scout: public misses and miss-target structure — information buying's public shadow). A noisy seat-vector that wanders from the quadrangle's shared centre must wander *in some direction*, and these two poles are the nearest distinct means in the two highest-variance observable directions; under a pure z-distance argmax, noise falls toward them. On this reading the attractor structure is a property of the observable space's geometry, not a defect of the model — but distinguishing those two explanations would need a calibration ablation this paper does not contain.

7 The fundamental limit, and the program this instrument belongs to

7.1 The divergence ceiling

The decision-divergence instrument of the v0.5 paper [1, 7] measures, by replaying identical views and seeds, the exact fraction of decisions on which each style differs from the Balanced control: Scout 2.89%, Ghost 2.15%, Archivist 1.66%, Banker 1.19%, Turtle 0.83%, Punter 0.47%, Hoarder 0.47%, Blitz 0.39% [7]. Those same measurements put the decision density at roughly 675 policy decisions per game across all six seats (declare-window decisions included), i.e. on the order of 110 per seat.

The arithmetic of the ceiling follows directly. A Blitz seat makes about 0.4 Blitz-revealing decisions per game; a Scout seat about 3. Every other decision is, by construction, *identical to the control's* — carrying zero information about the style. No observer of any channel, public or god's-eye, can classify a seat from decisions the seat's style never influenced; one game of a 0.39%-divergent style simply does not contain the evidence, and the measured quadrangle accuracy of 5.6–12.9% should be read as that fact, not as a classifier deficiency. The public channel then tightens the bound further: a share of the divergent decisions are declare-window choices, and a divergent *decline* emits no event at all (Section 2.3) — some of the few style-revealing decisions are invisible in principle. Styles are distinguishable in one game roughly in proportion to where their divergence lands: the Turtle's 0.83% is highly visible because it lands on asks and declares that reshape the whole public position, while the quadrangle's 0.39–1.19% lands on ask-scoring margins that the log renders as the same kind of event either way.

The constructive reading of the ceiling is about aggregation: at one game the observer is evidence-starved, not model-starved, so the natural upgrades are longitudinal — accumulating observations across games of a persistent opponent — rather than architectural. That is the direction the synthesis's program points, and it is future work.

7.2 The detection-panel program

The project's research synthesis [5] specifies (its §VI.4) an opponent-detection statistics panel — every statistic computable from the public log alone — as the feature set of its cross-game opponent-profiling style, "The Reader" (its S35). This paper's instrument is best understood as implementing and *measuring* a slice of that panel against a concrete roster. Shipped, with the panel's intended detection target: the dead-ask rate (its known-negative-ask detector), the certain-ask rate (its confirmation-ask / claim-imminent detector), ask diversity with hit rate (its signal-broker row), the same-book repeat rate (its chain-discipline row), the miss-target shares (the observable core of its turn-parking and targeting rows), declare timing via `declareBackload` (its claim-delay row, on the observable stand-in of Section 2.3), and the exact foreign / own-hand-only declare shares (its claimer taxonomy).

Unshipped, and honestly labelled future work: the blackballing detectors (collapse of a seat's inbound-target share); the evidence-age decay of hit rate — the synthesis's candidate best single skill estimator — which needs per-ask evidence dating; the possum detector (deducible certainties high while certain asks stay near zero), which requires running the full deduction engine over the log and is deliberately outside this observation layer's no-inference scope; claim-threshold estimation from provably contained books; and the entire convention-detection family — the Salahuddin-convention detector and the permutation-null excess-mutual-information test — which has nothing to detect here, since this roster plays no conventions, and which additionally needs revealed-hand data retained across games. None of the unshipped rows would lift the Section 7 ceiling for this roster in a single game; the panel's value concentrates in the cross-game, human-facing setting the synthesis designed it for.

8 Honest scope

Everything above is conditional, and the conditions are worth stating as bluntly as the results.

1. **This roster, this rule set.** Features were designed and fingerprints calibrated for the nine committed styles under `us54` as pinned by the rules hash [9, 6]. The feature set was written by authors who knew the roster — `foreignDeclareShare` exists because foreign-declaring styles do — so accuracy on a novel, adversarial, or human style is unmeasured, and the design leakage means it cannot be extrapolated from Table 4.

2. **Full-strength shared inference.** Every observed seat runs the identical full-strength inference engine [1]. Behavioural variation from *skill* — memory limits, deduction errors — is entirely absent from calibration. Human play varies along exactly that axis; it is out of distribution, and the synthesis's panel rows for skill estimation (evidence-age decay) are the missing instrument, not this one.

3. **Mirror calibration, cross-play deployment.** Fingerprints come from mirror games; accuracy is measured on cross-play, where game length and opponent behaviour differ. This shift is intentional (it is the deployment condition) but it means the reported accuracy is neither a ceiling nor a floor for other populations.

4. **Single-game observation only.** Nothing here aggregates across games. The divergence arithmetic of Section 7 says single-game reads of near-control styles are evidence-starved in principle; cross-game profiles are the synthesis's S35 program and remain future work.

5. **Deterministic policies.** All variation comes from seeds; a stochastic or deliberately deceptive policy (the synthesis's possum) faces detectors this layer deliberately does not include.

9 Conclusion

The `us54` public log is an unusually generous observation channel — it locates cards, certifies licences, and reveals declaration truth — and this paper measured how much play style survives the trip through it. The answer has a clean shape. What the channel certifies, it certifies exactly: foreign and own-hand-only declares are noiseless observations, and a 14-feature share vector over the log supports a simple, honest classifier once two measured corrections (no normaliser; half-pooled variances) are applied. What the channel recovers is governed by geometry that the roster's own companion findings predict: the four styles whose labelled axis is inert collapse to a point 0.20–0.25 wide and read at or near chance, while the styles whose divergence lands on public, position-reshaping events — Turtle, Hoarder, Scout, Ghost — are read at 27–50% against 11.1% chance. And what limits the whole enterprise is not the model but the material: styles that diverge on 0.39–2.89% of decisions offer an observer one to three revealing decisions per seat per game, some of them structurally invisible declines. Twice chance at one game is what this channel, this roster, and honest calibration yield; the road past it runs through cross-game aggregation, not a better single-game model. Whether even a perfect read would be worth anything to an adapting agent is the companion paper's question — and its answer belongs to that paper, not this one [2].

Reproducibility. The engine and classifier are deterministic. The committed fingerprint table carries its full provenance (generator command, seed prefix, per-bucket sample counts) [8]; the accuracy experiment's committed artifact carries the run configuration, health counters (all clean), records digest and engine commit [9]; and the distance matrix of Table 2 recomputes from the committed fingerprints alone.

10 Addendum: re-measured on the regenerated artifact

September 2026. Two things changed under this paper after it was written, and the classifier was re-calibrated and re-measured on both. The first is the concession layer of v2.0: every roster style now carries `defuse` = 1, which changes what the nine styles *do* and therefore what a public log about them contains. The second is a correction to `RULES_US54.md` §4, where a rule marked `[DERIVED]` advanced the turn to the next seat holding cards when a declare emptied the turn-holder — an opponent, where the owner's rulebook passes to a teammate [6]. The fingerprints of Section 4 were regenerated from fresh mirror games and the accuracy run of Section 6 re-played on the same 1,800 held-out cross-play games [9].

The headline is unchanged and the shape of the curve with it.

Twice chance and no better, at every horizon, on both runs. Nothing in Sections 7 or 8 moves: the ceiling is still the material rather than the model, and it is still set by how few decisions a style reveals.

Table 5: Single-game top-1 over nine styles, as published and as regenerated [9]. Chance is 11.1% throughout.

log horizon	as published	regenerated
40 events	16.4%	16.1%
80 events	20.2%	21.7%
full log	22.4%	22.0%

Table 6: End-of-game top-1 by style on the regenerated artifact, 1,200 seat-reads each [9]. The published run's two anchors were Ghost 50.4% and Balanced 5.6%.

style	top-1	style	top-1
Ghost	41.4%	Blitz	13.8%
Scout	32.7%	Banker	11.3%
Turtle	32.1%	Archivist	10.6%
Hoarder	29.3%	Punter	**3.6**%
Balanced	23.3%		

10.1 What did move: which style is hardest to read

The per-style column is a different matter, and this addendum will not smooth it over. As published, end-of-game accuracy ran from 50.4% on the Ghost down to 5.6% on the Balanced control, and the Balanced-is-invisible result was used in Section 7 as the sharpest illustration of the divergence argument. **On the regenerated artifact the floor belongs to a different style.**

The Ghost is still the most legible style and still by a wide margin. But Balanced has risen from 5.6% to 23.3% and **Punter has fallen to** 3.6%, so the hardest style to identify from a public log is now the roster's strongest one rather than its control.

Two readings are available and this paper commits to neither. The mechanical one is that the control is no longer the roster's midpoint: a mechanism every style inherits does not move a style relative to the others, but it does move where the *classifier's* decision boundaries fall, and Section 4.2's two calibration corrections were both fitted against a population that no longer exists. The substantive one is that Punter's behaviour has become the population's centre of mass, which would make it the style an unsure classifier defaults *away* from. **Distinguishing them needs a divergence measurement of the regenerated roster against its control, and that has not been run.**

10.2 What this addendum does not re-derive

The separability geometry of Section 5 — the cluster widths, the Turtle's distance from the inert group — and the two calibration corrections of Section 4.2 were not recomputed. They are statements about the feature space of the pre-concession roster, they are reported here as such, and the 40 of 60 control-game reads that forced the normaliser correction is a fact about that calibration rather than about this one. Anyone re-deriving them should expect the numbers to move by at least as much as Table 6 did.

References

[1] A. Hsieh. FishAI v0.5: Measuring play style without skill in a 54-card Literature variant. FishAI project; companion paper, published alongside this one, 2026.

[2] A. Hsieh. FishAI v1.0: When best-response adaptation is worth less than nothing. FishAI project; companion paper, 2026.

[3] A. Hsieh. The inert axis: when a style parameter is wired, swept, and never reached. FishAI project; companion paper, 2026.

[4] A. Hsieh. The contained book: an absorbing resource measured to be worth nothing. FishAI project; companion paper, 2026.

[5] FishAI project. PLAYSTYLES.md — Literature / Canadian Fish: play styles, rule dialects, and engine specification. Internal research synthesis with per-claim evidence tiers, 2026. §VI.4 specifies the opponent-detection statistics panel; S35 ("The Reader") the cross-game profiling program; S21 and S36 the possum and Salahuddin-convention detection targets.

[6] FishAI project. RULES_US54.md — the "US student" 54-card variant. Repository rules specification with derived consequences and test vectors, 2026. Rows 6–7 (ask licences), row 9 (hit transfer), and the declare-window mechanics cited in Section 2.

[7] FishAI project. STYLES.md — the play-style roster under `us54`. Repository document; §3.1.1 the Hoarder inertness audit, §6.1 the inert declare-threshold axis and per-style decision-divergence rates, 2026.

[8] FishAI project. Committed fingerprint calibration `lib/engine/bots/data/fingerprints.ts`. Generated by `node scripts/gen-fingerprints.mjs --games 150`: 150 `us54` mirror games per style, seed prefix `fingerprint-v1`, checkpoint buckets $\{60, 120, 200, 300, \texttt{full}\}$ with per-bucket sample provenance; generated 2026-08-24.

[9] FishAI project. Adaptive-lab artifact `src/lab/data/adaptive-results.json`. Classifier-accuracy block: 36 cross-play style pairings × 50 games (seed prefix `clsacc-v1`), truncation checkpoints $\{40, 80, 150, 250\}$ plus full log; records digest `ce3316641a8f38fe`; engine commit `45ec9f3` (recorded `45ec9f3-dirty`); health counters all zero; generated 2026-08-24.

[10] FishAI project. FishAI repository: engine, bots, simulation laboratory, and results site. `lib/engine/bots/observe.ts` (the observation layer and its header derivations), `lib/engine/bots/classify.ts` (the classifier and the two calibration corrections), `tests/bots/observe.test.ts` and `tests/bots/classify.test.ts` (the pinned contracts), 2026.